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RETENTION ASSEMBLIES FOR MATERIAL STORAGE SYSTEMS, AND RELATED MATERIAL CONTAINERS AND METHODS

Abstract

The present disclosure relates to retention assemblies for material storage systems, and related material containers and methods. In one or more embodiments, a retention assembly includes an actuator mounted to a retention housing, and a locking component coupled to the actuator. The actuator is movable to move the locking component between a locked position and an unlocked position. The retention assembly includes a magnet coupled to the actuator.

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Background/Summary

INTRODUCTION

[0001] Material storage containers can be limited with respect to modularity and use. For example, it can be difficult and time consuming to refill storage containers because the storage containers can be heavy and large. Moreover, storage containers can be difficult to retain in place to prevent movement of the storage containers.

SUMMARY

[0002] The present disclosure relates to retention assemblies for material storage systems, and related material containers and methods.

[0003] In one or more embodiments, a retention assembly includes an actuator mounted to a retention housing, and a locking component coupled to the actuator. The actuator is movable to move the locking component between a locked position and an unlocked position. The retention assembly includes a magnet coupled to the actuator.

[0004] In one or more embodiments, a material container includes a storage volume defined at least partially by a storage housing, and a nozzle pivotably coupled to the storage housing. The nozzle includes a flow housing at least partially defining an outlet opening, and a shaft coupled to the flow housing and extending into a section of the storage housing. The shaft includes a flow opening between the storage volume and the outlet opening.

[0005] In one or more embodiments, a method of using a material storage system includes moving an actuator of a retention assembly to move a locking component of the retention assembly to an unlocked position to unlock a material container retained by the retention assembly. The moving of the actuator at least partially aligns a magnet coupled to the actuator with a second magnet coupled to the material container. The method includes lifting a material container off of the retention assembly, moving the material container away from a vehicle, and dispensing a material from the material container.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] So that the manner in which the above recited features of the present disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only exemplary embodiments and are therefore not to be considered limiting in scope, and may admit to other equally effective embodiments.

[0007] FIG. **1** is a schematic perspective view of a vehicle, according to one or more embodiments.

[0008] FIG. **2** is a schematic axonometric view of a first side of a material storage system, according to one or more embodiments.

[0009] FIG. **3** is a schematic axonometric view of a second side of the material storage system shown in FIG. **2**, according to one or more embodiments.

[0010] FIG. **4** is a schematic partial cross-sectional view of the material storage system along Section **4-4** shown in FIG. **2**, according to one or more embodiments.

[0011] FIG. **5** is a schematic partial bottom view of the retention assembly shown in FIGS. **2-4** with the base section hidden, according to one or more embodiments.

[0012] FIG. **6** is a schematic partial top view of the material container shown in FIGS. **2-5** with the storage housing hidden, according to one or more embodiments.

[0013] FIG. **7** is a schematic partial side view of the material storage system shown in FIGS. **1-6**,

according to one or more embodiments.
[0014] FIG. **8** is a schematic partial axonometric view of a nozzle, according to one or more embodiments.
[0015] FIG. **9** is a schematic cross-sectional view of the nozzle along Section **9-9** shown in FIG. **8**, according to one or more embodiments.
[0016] FIGS. **10**A-**10**D illustrate a schematic process flow of a method of using the material storage system, according to one or more embodiments.
[0017] FIG. **11** is a schematic partial side view of the retention assembly shown in FIGS. **2-10**, according to one or more embodiments.
[0018] FIG. **12** is a schematic partial back view of a storage container coupled to a vehicle, according to one or more embodiments.
[0019] FIG. **13** is a schematic partial back view of the storage container shown in FIG. **12** with the one or more doors open, according to one or more embodiments.
[0020] To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. It is contemplated that elements and features of one embodiment may be beneficially incorporated in other embodiments without further recitation.

DETAILED DESCRIPTION

[0021] The present disclosure relates to retention assemblies for material storage systems, and related material containers and methods.
[0022] In one or more embodiments, a material storage system includes a material container and a retention assembly. The material container includes a storage volume defined at least partially by a storage housing. The retention assembly is positionable to interface with the material container. The retention assembly includes an actuator mounted to a retention housing, and a locking component coupled to the actuator. The actuator is movable to move the locking component to lock and unlock the material container to and from the retention assembly. The retention assembly is operable to retain the material container when the material container is locked to the retention assembly. Additionally, the retention assembly includes a magnet coupled to the actuator, and the material container includes one or more second magnets.
[0023] The material storage system can be filled with fluid (such as liquid and/or gas), solids, and/or a combination (such as a slurry) thereof. The material storage system can store other materials. The material storage system can be connected to a spray nozzle. The fluid storage system can be used, for example, to spray off equipment and/or people (such as for cleaning and/or showering during camping), and/or to provide drinking water and/or cooking water. As described above, materials other than water may be in the material storage system. In one or more embodiments, a nozzle (e.g., a passive nozzle) is pivotably coupled to the storage housing. The nozzle includes a flow housing at least partially defining an outlet opening, and a shaft coupled to the flow housing and extending into a section of the storage housing. The shaft includes a flow opening between the storage volume and the outlet opening.
[0024] In operation, the material container can be lowered onto the retention assembly such that the material container is locked to the retention assembly using the locking component. The material container can be unlocked from the retention assembly by moving the actuator to an unlocked position. In the unlocked position, the magnet coupled to the actuator aligns with at least one of the one or more second magnets included in the material container, such that a magnetic force (e.g., an attractive force) between the magnet and the second magnet(s) maintains the actuator in the unlocked position when the actuator is released. Hence, the material container can be lifted off of the retention assembly without needing to continuously hold the actuator in the unlocked position.
[0025] The material storage system and retention assembly enable quick and easy locking and locking of a material container to and from the retention assembly. For example, the actuator can be manipulated to the unlocked position and can remove the material container from the retention

assembly after releasing the actuator. Moreover, the material container can be locked to the retention assembly by lowering the material container onto the retention assembly without manipulation of the actuator. The present disclosure contemplates that the operations described herein (such as the lowering, the lifting, the unlocking, and/or the manipulation) can be conducted manually by a user and/or automatically using automated equipment.

[0026] The disclosure contemplates that terms used herein such as “couples,” “coupling,” “couple,” and “coupled” may include but are not limited to bonding, embedding, welding, fusing, melting together, interference fitting, interference abutting, and/or fastening such as by using bolts, threaded connections, pins, and/or screws. The disclosure contemplates that terms such as “couples,” “coupling,” “couple,” and “coupled” may include but are not limited to integrally forming. The disclosure contemplates that terms such as “couples,” “coupling,” “couple,” and “coupled” may include but are not limited to direct coupling and/or indirect coupling, such as indirect coupling through components such as links, blocks, and/or frames.

[0027] FIG. **1** is a schematic perspective view of a vehicle **100**, according to one or more embodiments. The vehicle **100** may include multiple sensors **101** and/or multiple cameras **102**. The vehicle **100** includes one or more wheel modules **107**. The vehicle **100** is an automotive vehicle. In the implementation shown in FIG. **1**, the vehicle **100** is a truck, such as an electric truck. The present disclosure contemplates that the subject matter described herein can be used in any other type of vehicle having any numbers of wheels, such as vans, internal combustion vehicles, and/or sport-utility vehicles (SUVs). The vehicle **100** includes a view panel **111** mounted to a vehicle body **108** of the vehicle **100**.

[0028] FIG. **2** is a schematic axonometric view of a first side of a material storage system **200**, according to one or more embodiments. The material storage system **200** can be supported in or on a vehicle (such as the vehicle **100** shown in FIG. **1**). The material storage system **200** can be supported in or on, for example, a bed, a cab, a hood, a trunk, a gear tunnel, a roof, and/or a floor of a vehicle. As another example, the material storage system **200** can be supported in or on a storage container that is supported by and/or coupled to a vehicle.

[0029] The material storage system **200** includes a material container **201** and a retention assembly **230**. The material container **201** includes a storage housing **202** and a carry handle **203** coupled to the storage housing **202**. FIG. **2** shows the material container **201** as locked to the retention assembly **230**. The material container **201** can be locked to and unlocked from the retention assembly **230**. A spray nozzle **204** and a ring **205** are supported in a recess of the storage housing **202**. A hose is connected to the spray nozzle **204** and a connector **206**. The hose is wound about the ring **205**. A nozzle **280** is pivotably coupled to the storage housing **202**. The nozzle **280** is operable to flow a material in the material container **201**. The nozzle **280** is shown as disposed above the nozzle **245** of the retention assembly **230**. The nozzle **280** of the material container **201** can be moved to other positions (such as above the actuator **231**). The material container **201** includes a removable cap **207** coupled to the storage housing **202**. The removable cap **207** can be removed such that the material container **201** can be at least partially filled with the material.

[0030] The retention assembly **230** includes an actuator **231** mounted to a retention housing **232**. FIG. **2** shows the actuator **231** in a locked position that locks the material container **201** to the retention assembly **230**.

[0031] FIG. **3** is a schematic axonometric view of a second side of the material storage system **200** shown in FIG. **2**, according to one or more embodiments.

[0032] The retention assembly **230** includes a nozzle **245** coupled to the retention housing **232**, a power button **246**, one or more electrical indicators **247**, and an electrical port **248**. The electrical port **248** can transmit data and/or electrical power to and from the retention assembly **230**. As an example, the electrical port **248** can recharge the battery **263** described below. As another example, the electrical port **248** can transmit power from the battery **263** and to a mobile device (or other equipment) such that the retention assembly **230** can be used as a power bank. The electrical port

248 can be a USB-C port. Other ports are contemplated. The retention assembly **230** can be integrated with a vehicle (such as the vehicle **100**) such that electrical power and/or data can be supplied from the power system of the vehicle and to the retention assembly **230** through the electrical port **248**. The nozzle **245** is operable to flow the material in the material container **201**. In one or more embodiments, the nozzle **245** of the retention assembly **230** is an active nozzle, and the nozzle **280** of the material container **201** is a passive nozzle.

[0033] FIG. **4** is a schematic partial cross-sectional view of the material storage system **200** along Section **4-4** shown in FIG. **2**, according to one or more embodiments.

[0034] The retention assembly **230** includes a locking component **233** and a second locking component **234** coupled to the actuator **231**. The actuator **231** is movable to move the locking component **233** and the second locking component **234** between the locked position and an unlocked position. The retention assembly **230** includes a rotatable link **235** pivotably coupled to the locking component **233** and the second locking component **234**. In one or more embodiments, the retention housing **232** includes a base section **232***a* and a lid section **232***b* coupled together. The material container **201** includes an extension **208** extending relative to the storage housing **202**, an opening **210** formed in the extension **208**, and a second opening **211** formed in the extension **208**. The actuator **231** is movable to move the locking component **233** into and out of the opening **210**. The actuator **231** is also movable to move the second locking component **234** into and out of the second opening **211**. FIG. **4** shows the locking components **233**, **234** in the locked position with the locking components **233**, **234** extending respectively into the openings **209**, **210**. The actuator **231** can be moved to the unlocked position such that the locking components **233**, **234** are retracted respectively out of the openings **209**, **210**. The storage housing **202** at least partially defines a storage volume **214** that is at least partially filled with the material.

[0035] The retention assembly **230** includes a column **241** disposed in a conduit **238**. The conduit **238** is fluidly connected to the nozzle **245** (shown in FIG. **3**) of the retention assembly **230**. When the material container **201** is lowered onto the retention assembly **230**, the extension **208** pushes on angled (e.g., tapered) or curved outer surfaces of the locking components **233**, **234** to push the locking components **233**, **234** toward the rotatable link **235** until the locking components **233**, **234** can move into the openings **210**, **211** and into the locked position. Also, when the material container **201** is lowered, the column **241** pushes a plunger **212** against a bias element **213** to move the plunger **212** upward. The movement of the plunger **212** allows the material in the material container **201** to flow past the plunger **213**, through the conduit **238**, and out through the nozzle **245**.

[0036] In one or more embodiments, the actuator **231** includes a toggle button, the locking component **233** includes a first striker, the second locking component **234** includes a second striker, and/or the rotatable link **235** includes a pawl. FIG. **4** shows the nozzle **280** schematically and without hatching.

[0037] FIG. **5** is a schematic partial bottom view of the retention assembly **230** shown in FIGS. **2-4** with the base section **232***a* hidden, according to one or more embodiments. The retention assembly **230** includes a magnet **249** coupled to the actuator **231**, and a transfer link **236** coupled between the rotatable link **235** and the actuator **231**. The rotatable link **235** is rotatable relative to the retention housing **232** and the storage housing **202**. The transfer link **236** is pivotable relative to the rotatable link **235** and the actuator **231** such that when the actuator **231** moves the transfer link **236** pivots to rotate the rotatable link **235**. The rotation of the rotatable link **235** moves the locking components **233**, **234**. In one or more embodiments, the locking components **233**, **234** are plates. The rotatable link **235** includes an extension **251** extending into an opening **252** of the transfer link **236**.

[0038] First extensions **250** of the lid section **232***b* extend into channels formed in the locking components **233**, **234** to guide the locking components **233**, **234** as the locking components **233**, **234** extend toward the locked position and retract toward the unlocked position. The retention assembly **230** includes a bias element **237** operable to bias the actuator **231** toward the locked position. In one or more embodiments, the bias elements **213**, **237** respectively include a spring, such as a coil spring, for example a helical coil spring. The bias elements **213**, **237** can respectively be tape springs, wave springs, torsion springs, and/or flat springs. Other springs and/or other bias elements are contemplated. In one or more embodiments, the bias element **237** is under compression and expands to bias the actuator **231** to the locked position. The bias element **237** is coupled to the extension **251** of the rotatable link **235** and a second extension **276** of the lid section **232***b*. The magnet **249** is configured resist the bias element **237** to hold the actuator **231** in the unlocked position when an accessory (such as the material container **201**) is positioned on the retention assembly **230**.

[0039] The locking component **233** interfaces with a first side of the rotatable link **235**, and the second locking component **234** interfaces with a second side of the rotatable link **235**, the second side opposing the first side. The rotatable link **235** includes a second extension **253** extending into an opening **254** of the locking component **233**, and a third extension **255** extending into a second opening **256** of the second locking component **234**.

[0040] The actuator **231** includes a handle **257** and a receptacle **258** having an opening **259**. The magnet **249** is disposed in the opening **259**. In one or more embodiments, the magnet **249** is bonded (e.g., glued) to the actuator **231**. The actuator **231** is coupled to the transfer link **236** using a flange **260** abutting against a protrusion **269** of the transfer link **236**. The retention assembly **230** includes a pump **261** (shown in ghost for visual clarity purposes) fluidly connected to the nozzle **245**, a controller **262** in communication with the pump **261**, and a battery **263** (shown in ghost for visual clarity purposes) electrically connected to the controller **262**. The controller **262** can include, for example, a memory and/or a processor. The controller **262** can include a printed circuit board (PCB). The pump **261** is an electric pump **261**. The pump **261** pressurizes the material to pump the material through the nozzle **245** and the spray nozzle **204**. The material storage system **200** can include one or more heat transfer elements **265** operable to heat and/or cool the material. The one or more heat transfer elements **265** can, for example, be coupled to the retention assembly **230** (such as coupled to the pump **261**) to heat and/or cool the material as the material is dispensed and/or be coupled to the material container **201** to heat and/or cool the material as the material is stored in the material container **201**.

[0041] The material container **201** includes one or more second magnets **215** (a plurality, e.g., four, are shown). As the actuator **231** moves (e.g., along a first direction D**1**) to the locked position, the magnet **249** aligns under one of the second magnets **215** to generate and/or increase an attractive magnetic force between the magnet **249** and the respective second magnet **215**. In one or more embodiments, the magnet **249** and the one or more second magnets **215** are ferromagnetic. The attractive magnetic force is sufficient to overcome a bias force of the bias element **237** to maintain the actuator **231** in the locked position after the handle **257** is released by a user and/or automated equipment, which allows the material container **201** to be lifted off of the retention assembly **230** without continued manipulation of the handle **257**. During the lifting of the material container **201**, the respective second magnet **215** is moved away from the magnet **249** by a sufficient distance to reduce or eliminate the attractive magnetic force. When the attractive magnetic force is reduced or eliminated, the bias force of the bias element **237** moves the actuator **231** (e.g., along a second direction D**2**) back to the locked position shown in FIG. **5**. The bias element **237** biases the locking component **233** toward the opening **210** (see FIG. **4**) of the material container **201** and biases the second locking component **234** toward the second opening **211** (see FIG. **4**) of the material container **201**.

[0042] FIG. **6** is a schematic partial top view of the material container **201** shown in FIGS. **2-5** with the storage housing **202** hidden, according to one or more embodiments.

[0043] The one or more second magnets **215** are respectively disposed in opening(s) **267** of one or more protrusions **266** of the extension **208**. In one or more embodiments, the respective second magnets **215** are bonded (e.g., glued) to the one or more protrusions **266**. The second magnets **215** are spaced from each other along an azimuthal direction AD**1**. The second magnets **215** are spaced equidistantly from each other by an azimuthal angle A**1** such that, if the material container **201** is rotated, the magnet **249** aligns with at least one of the one or more second magnets **215** when in the locked position. In one or more embodiments, the azimuthal angle A**1** is about 90 degrees.

[0044] FIG. **7** is a schematic partial side view of the material storage system **200** shown in FIGS. **1-6**, according to one or more embodiments.

[0045] FIG. **7** shows the actuator **231** in solid in the locked position, and in ghost in the unlocked position. The magnet **249** (shown in ghost) moves between a first position **749***a* in the locked position and a second position **749***b* in the unlocked position. In the locked position, the magnet **249** is offset from the respective second magnets **215**. In the unlocked position, the magnet **249** is at least partially aligned under at least one of the second magnets **215**.

[0046] FIG. **8** is a schematic partial axonometric view of a nozzle **880**, according to one or more embodiments. The nozzle **880** can be used in place of the nozzle **280** of the material container **201** shown in FIGS. **2** and **4**.

[0047] FIG. **2** shows the nozzle **280** in a collapsed position, and FIG. **8** shows the nozzle **880** pivoted into an extended position. The nozzle **880** includes a flow housing **881** at least partially defining an outlet opening **882**. The nozzle **880** includes a shaft **883** coupled to the flow housing **881** and extending into a section **287** of the storage housing **202**. In one or more embodiments, the shaft **883** extends through the section **887** of the storage housing **202** and into the storage volume **214**. The shaft **883** includes a flow opening **884** between the storage volume **214** and the outlet opening **882**. A washer seal **885** is disposed about the shaft **883** of the nozzle **880**. The shaft **883** is rotatable relative to the storage housing. The nozzle **880** includes a valve **888** coupled to the flow housing **881**. The valve **888** can be, for example, an air return valve. Other types of valves are contemplated.

[0048] FIG. **9** is a schematic cross-sectional view of the nozzle **880** along Section **9-9** shown in FIG. **8**, according to one or more embodiments.

[0049] In one or more embodiments, the shaft **883** is threaded into the flow housing **881** and is disposed about a rod section **889** of the flow housing **881**. An opening **890** of the rod section **889** fluidly connects the flow opening **884** of the shaft **883** to the outlet opening **882**. A second washer seal **891** is disposed about the rod section **289**. The rod section **889**, the shaft **883**, and the flow housing **881** can move (e.g., rotate) together. The rod section **889** can be integrally formed with the flow housing **881** or can inserted into (e.g., threaded into) the flow housing **881**. The second washer seal **891** is disposed between the flow housing **881** and an end of the shaft **883**.

[0050] FIGS. **10**A-**10**D illustrate a schematic process flow of a method of using the material storage system **200**, according to one or more embodiments.

[0051] The description of the method includes references to reference numerals of the material storage system **200**, and the present disclosure contemplates that subject matter (such as structures and components) other than the subject matter of the material storage system **200** can be used in relation to the method.

[0052] At FIG. **10**A, the retention assembly **230** is shown with the actuator **231** in the locked position. The material container **201** is positioned in relation to (e.g., above) the retention assembly **230**.

[0053] At FIG. **10**B, the material container **201** is positioned (e.g., lowered) to interface with the retention assembly **230** to lock the material container **201** to the retention assembly **230**.

[0054] At FIG. **10**C, the method includes moving the actuator **231** to move the locking component **233** and the second locking component **234** to the unlocked position to unlock the material container **201** retained by the retention assembly **230**. The moving of the actuator **231** at least partially aligns the magnet **249** coupled to the actuator **231** with at least one of the second magnet **215** coupled to the material container **201**.

[0055] At FIG. **10**D, the material container **201** is lifted off of the retention assembly **230**. The method can include moving the material container **201** away from a vehicle (such as the vehicle **100**). The lifting of the material container **201** moves the at least one second magnet **215** away from the magnet **249** to bias the actuator **231** to the locked position. FIG. **10**D shows the actuator **231** as the actuator **231** is automatically biased toward the locked position.

[0056] The method can include dispensing the material from the material container **201**. For example, the material can be dispensed using the nozzle **280** and/or the nozzle **880** of the material container **201**. As another example, the material can be dispensed using the nozzle **245** of the retention assembly **230** when the material container **201** is interfacing with the retention assembly **230** (e.g., in FIG. **10**B and/or FIG. **10**C). The method can include connecting (e.g., using the connector **206**) the spray nozzle **204** to the nozzle **245** of the retention assembly **230** prior to the dispensing of the material.

[0057] In one or more embodiments, the method includes at least partially filling the storage volume **214** of the material container **201** with the material before FIG. **10**A and/or after FIG. **10**D when the material container **201** is lifted off of the retention assembly **230**. After the storage volume **214** is at least partially filled, the material container **201** can be lowered and/or re-lowered (as shown in FIG. **10**B) onto the retention assembly **230** to lock the material container **201** to the retention assembly **230**.

[0058] When the fill level of the material container **201** is low, the material container **201** can be removed from the retention assembly **230** (as shown in FIG. **10**D), and a second material container **201** (e.g., having a higher fill level) can be lowered and locked to the retention assembly **230** (as shown in FIG. **10**B).

[0059] FIG. **11** is a schematic partial side view of the retention assembly **230** shown in FIGS. **2-10**, according to one or more embodiments.

[0060] The retention assembly **230** is coupled to a vehicle panel **1100** of a vehicle. The retention assembly **230** includes an extension **270** including an outer surface **271** and at least one opening **272** formed in the outer surface **271**. The present disclosure contemplates that the extension **270** can be separated into a plurality of extensions. The retention assembly **230** includes at least one second actuator **273** mounted to the retention housing **232**, and at least one third locking component **274** disposed in the at least one opening **272** and coupled to the at least one second actuator **273**.

[0061] The at least one second actuator **273** is movable to move the at least one third locking component **274** between a first position (shown in solid in FIG. **11**) and a second position (shown in ghost in FIG. **11**). In the first position, the at least one third locking component **274** is at least partially outside of the at least one opening **272** and abutting against the vehicle panel **1100**. In the second position the at least one third locking component **274** is moved into the at least one opening **272** relative to the first position. The implementation in FIG. **11** shows two third locking components **274** and two second actuators **273**. The respective second actuators **273** are rotatable to pivot the respective third locking components **274** between the first position and the second position. In the first position, the third locking components **274** extend into one or more openings of the vehicle panel **1100** to lock the retention assembly **230** to the vehicle panel **1100**. The second actuators **273** can include cams having key slots such that a key can be used to turn the second actuators **273** and lock the third locking components **274** in the first position or the second position. The present disclosure contemplates that the second actuators **273** can include pivotable bars. The vehicle panel **1100** can be disposed along a bed, a cab, a hood, a trunk, a gear tunnel, a roof, and/or a floor of a vehicle. As another example, the vehicle panel **1100** can be disposed along a storage container that is supported by and/or coupled to a vehicle.

[0062] The retention assembly **230** can be movable relative to a vehicle (as shown in FIGS. **2-10**) or locked in place relative to the vehicle (as shown in FIG. **11**). The retention housing **232** can be integrally formed with the vehicle panel **1100**.

[0063] FIG. **12** is a schematic partial back view of a storage container **1200** coupled to a vehicle

1250, according to one or more embodiments. The storage container **1200** can be attached, for example, to a rear bumper **1251** of the vehicle **1250**. The vehicle **1250** is shown as an SUV. Other vehicles are contemplated.

[0064] The storage container **1200** includes one or more doors **1201**, **1202** (two are shown) a roof **1203**, and a bench **1204**. The bench **1204** can include a cook top **1205** (such as a stove top). The bench **1204** can be omitted and/or the roof **1203** can be moved to provide access inside of the storage container **1200**. The one or more doors **1201**, **1202** can be opened e.g., pivoted) to provide access inside of the storage container **1200**. The present disclosure contemplates that the one or more doors **1201**, **1202** can be replaced with stationary wall(s). One or more support bars can be coupled to the storage container **1200** and the vehicle **1250** such that the vehicle **1250** supports the storage container **1200**. The one or more support bars can extend at least partially between the vehicle **1250** to space the storage container **1200** from the vehicle **1250**.

[0065] FIG. **13** is a schematic partial back view of the storage container **1200** shown in FIG. **12** with the one or more doors **1201**, **1202** open, according to one or more embodiments.

[0066] A cooler **1206**, a trash container **1207**, an equipment container **1208** (containing, e.g., pots and pans and/or kitchen utensils), and/or the material storage system **200** can be stored in the storage container **1200**. The doors **1201** can also hold equipment (such as cooking ingredients, kitchen utensils, and/or other equipment). The retention assembly **230** is coupled to a floor **1209** of the storage container **1200**. The retention housing **232** can be integrally formed floor **1209**. The vehicle panel **1100** described in relation to FIG. **11** can be part of the floor **1209**, and the retention assembly **230** can be removably locked to the floor **1209** in a manner—for example—described in relation to FIG. **11**. Although not shown in FIG. **13**, the present disclosure contemplates that the actuator **231**, the nozzle **245**, the nozzle **280** (and/or the nozzle **880**), the power button **246**, the one or more electrical indicators **247**, and/or the electrical port **248** can be positioned for easy access (as an example, facing door **1202**) when the material container **201** is locked to the retention assembly **230** inside of the storage container **1200**. Other positions and/or orientations are contemplated for the actuator **231**, the nozzle **245**, the nozzle **280** (and/or the nozzle **880**), the power button **246**, the one or more electrical indicators **247**, and/or the electrical port **248**.

[0067] Benefits of the present disclosure include but are not limited to ease and simplicity of mechanically locking and automatically mechanically (e.g., magnetically) unlocking of material containers to and from retention assemblies; smaller and/or lighter material containers for easy and quick refilling; and reliable retention of storage containers to reduce or eliminate movement of the storage containers.

[0068] It is contemplated that one or more aspects disclosed herein may be combined. As an example, one or more aspects, features, components, operations and/or properties of the vehicle **100**, the material container **201**, the retention assembly **230**, the nozzle **280**, nozzle **880**, the method shown in FIGS. **10**A-**10**D, the retention assembly **230** implementation shown in FIG. **11**, the vehicle panel **1100**, and/or the storage container **1200** may be combined. Moreover, it is contemplated that one or more aspects disclosed herein may include some or all of the aforementioned benefits.

[0069] While the foregoing is directed to embodiments of the present disclosure, other and further embodiments of the disclosure may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

Claims

1. A retention assembly, comprising: an actuator mounted to a retention housing; a locking component coupled to the actuator, the actuator movable to move the locking component between a locked position and an unlocked position; and a magnet coupled to the actuator.

2. The retention assembly of claim 1, further comprising a bias element operable to bias the

actuator toward the locked position, wherein the magnet is configured resist the bias element to hold the actuator in the unlocked position when an accessory is positioned on the retention assembly.
3. The retention assembly of claim 1, further comprising: a rotatable link pivotably coupled to the locking component; and a transfer link coupled between the rotatable link and the actuator.
4. The retention assembly of claim 3, wherein the rotatable link is rotatable relative to the retention housing, and the transfer link is pivotable relative to the rotatable link and the actuator.
5. The retention assembly of claim 3, wherein the rotatable link comprises an extension extending into an opening of the transfer link.
6. The retention assembly of claim 3, wherein: the locking component interfaces with a first side of the rotatable link; the retention assembly further comprises a second locking component interfacing with a second side of the rotatable link, the second side opposing the first side; and the rotatable link comprises an extension extending into an opening of the locking component.
7. The retention assembly of claim 3, wherein the actuator comprises: a handle; a receptacle having an opening, wherein the magnet is disposed in the opening; and a flange abutting against the transfer link.
8. The retention assembly of claim 1, wherein the retention assembly comprises: an extension comprising an outer surface and an opening formed in the outer surface; a second actuator mounted to the retention housing; and a second locking component disposed in the opening and coupled to the second actuator.
9. The retention assembly of claim 8, wherein the second actuator is movable to move the second locking component between a first position in the opening and a second position, wherein in the second position the second locking component is at least partially outside of the opening and abutting against a vehicle panel.
10. A material storage system, comprising: a material container comprising a storage volume defined at least partially by a storage housing; and the retention assembly of claim 1, wherein the material container is positionable to interface with the retention assembly, and the locking component moves between the locked position and the unlocked positon to lock and unlock the material container to and from the retention assembly.
11. The material storage system of claim 10, wherein: the material container further comprises: one or more second magnets, an extension extending relative to the storage housing, and an opening formed in the extension, wherein the actuator is movable to move the locking component into and out of the opening; and the retention assembly further comprises: a bias element operable to bias the locking component toward the opening.
12. The material storage system of claim 10, wherein the retention assembly further comprises: a nozzle operable to flow a material; a pump fluidly connected to the nozzle; a controller in communication with the pump; and a battery electrically connected to the controller.
13. A material container, comprising: a storage volume defined at least partially by a storage housing; and a nozzle pivotably coupled to the storage housing, the nozzle comprising: a flow housing at least partially defining an outlet opening, and a shaft coupled to the flow housing and extending into a section of the storage housing, the shaft comprising a flow opening between the storage volume and the outlet opening.
14. The material container of claim 13, further comprising a washer seal disposed about the shaft of the nozzle, wherein the shaft is rotatable relative to the storage housing.
15. The material container of claim 13, wherein the nozzle further comprises a valve coupled to the flow housing.
16. A method of using a material storage system, comprising: moving an actuator of a retention assembly to move a locking component of the retention assembly to an unlocked position to unlock a material container retained by the retention assembly, the moving of the actuator at least partially aligning a magnet coupled to the actuator with a second magnet coupled to the material container;

lifting a material container off of the retention assembly; moving the material container away from a vehicle; and dispensing a material from the material container.

17. The method of claim 16, wherein the retention assembly is coupled to a vehicle panel of the vehicle.

18. The method of claim 16, wherein the lifting of the material container moves the second magnet away from the magnet to bias the actuator to a locked position, and the method further comprises: at least partially filling a storage volume of the material container with the material.

19. The method of claim 18, further comprising lowering the material container onto the retention assembly to lock the material container to the retention assembly.

20. The method of claim 16, further comprising connecting a spray nozzle to a nozzle of the retention assembly.